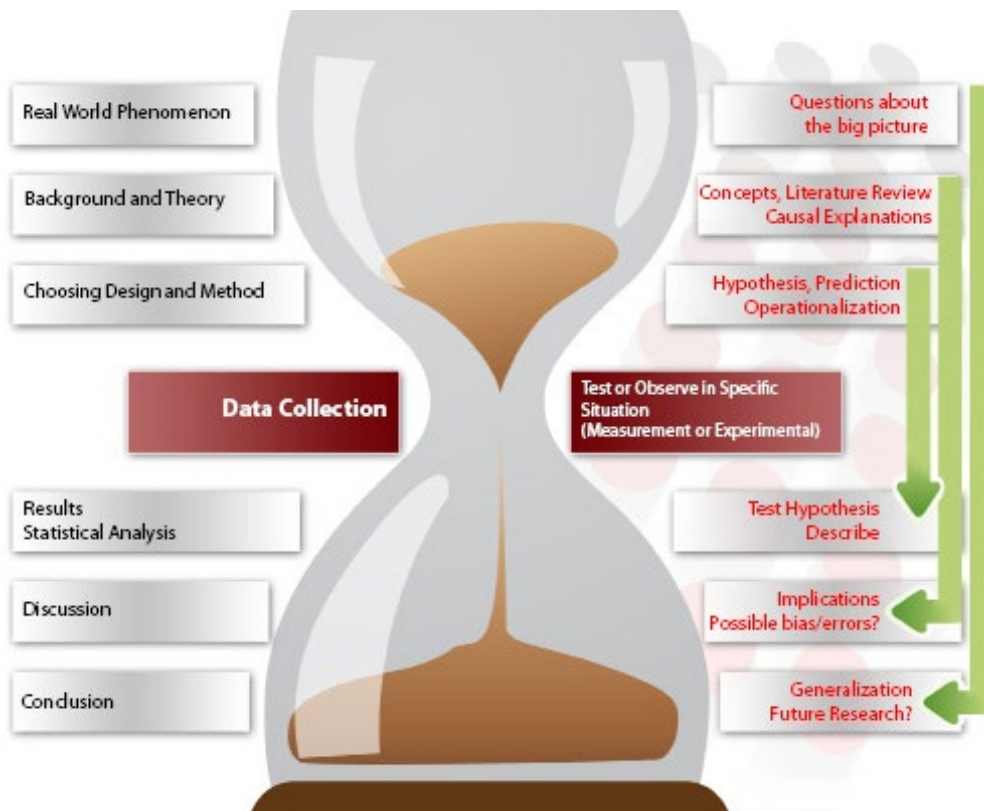




Study Guide for MGTE 43216 / INTE 43216 - Research Project



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Syllabus

Course Code : MGTE 43216 / INTE 43216
Course Title : Research Project
Prerequisites : MGTE 31222 - Research Methods

Learning outcomes:

On completion of this course, the student should be able to:

- explain the scientific process of carrying out research
- identify and formulate a suitable research problem in management and information technology areas
- design and carry out experiments, data collection and analysis
- critically analyse the research results
- summarize the results obtained and write a research paper

Course content:

Each student is assigned a project by the department in areas such as finance, operations, marketing, information systems, software engineering etc. The projects assigned are directly relevant to a typical industry issue and the students are expected to apply the knowledge gained throughout the programme.

Method of teaching and learning:

Progress monitoring continuously by a series of seminars, presentations, discussions, and progress review meetings.

Assessment:

End-of-the year presentation and dissertation evaluation, mid-year assessments, continuous assessment.

Recommended reading:

1. Colin Robson, How to do a Research Project - A Guide for Undergraduate Students, 2nd Edition, John Wiley & Sons Publishing, 2016.
2. Study Guide for MGTE 43216 / INTE 43216 - Research Project, Department of Industrial Management, Faculty of Science, University of Kelaniya, 2017.

1. Introduction

As a student reads for B.Sc. (Honours) in Management and Information Technology (MIT) / B.Sc. (Honours) in Information Technology (IT) degree, you are required to successfully complete a research-based project in partial fulfilment of the requirement for the award of the degree. This research project carries a weight of six (6) credits and is a core course unit which follows the supportive course unit on research methodology. The main objective of this research project is to provide an opportunity for you to apply conceptual learning to the practice of management and information technology. Moreover, you will get hands-on experience on how to organize and carry out a research project independently while interacting with other stakeholders including experienced industry professionals. You will also be able to make a positive contribution to existing knowledge as your research findings may benefit the industry, fellow researchers, academia and yourself.

You may begin your research project by identifying a problem related to industrial / business application or on a fundamental concept / theory. You are advised to identify a problem and then consult a supervisor appointed by the Department. From this point onwards, you will go through the following chronological events. Timing and deadlines for each event is stipulated and announced by the coordinator of the research projects as per the schedule given in Annex I.

1. Supervisor approves the research problem and the problem domain as suitable for an undergraduate research project.
2. Supervisor requests you to make a formal research proposal of which the format is given in Annex II.
3. You are required to network with other stakeholders such as personnel in the industry involved to get necessary assistance if needed.
4. You then carry out the research work, constantly seeking guidance and feedback from the supervisor. You are responsible for meeting the supervisor on a regular basis. Your supervisor and you are required to maintain a log that shows details of student-supervisor interaction.
5. The Department arranges for mid-year project presentations by students. The aim of mid-year presentation is to monitor progress and to provide feedback from an extended panel of evaluators and bring another perspective to the problem being researched. This step will enable you to receive guidance from experts other than your supervisor.
6. You complete a survey of literature, formulate a scientific methodology to solve the identified problem, collect and collate data, develop any prototype (if applicable), analyse and interpret data, discuss your findings and critically evaluate your own work and provide initial conclusions.

This will be presented through a preliminary dissertation to the supervisor, who may approve without or with minor/major changes, or reject it as not suitable for the final presentation. If it is accepted, you will make a formal final presentation before a panel of examiners who will evaluate and award a final grade for this phase. Your final grade will be evaluated based on the assessments made throughout the project period. Evaluation criterion of your research project is given in Annex III.

If supervisor deems that your research work is not suitable for the final presentation or if you are unable to meet the deadline, you will not be allowed to do the final presentation and you will have to be considered as a repeat student.

7. Once you have presented your research work before a panel of examiners, you are then required to submit two copies of the hard-bound final dissertation to the Department. The structure of final dissertation is given in Annex IV.

2. Beginning your research project

You must identify an *issue* related to an industrial / business activity or fundamental concept and transform it to a *researchable problem*. Once you identify such an issue, reference or literature survey will help you to concretise the issue and define your research problem. Once the research problem and its domain have been identified, the student must clearly define research goals, objectives and outcomes. It should be noted that the initial objectives in most occasions will get fine-tuned based on literature review and/or preliminary exploratory study on the problem and the current state of knowledge on it.

2.1 Define the purpose

The purpose of research explicitly explains what you will achieve in terms of defined and specific benefits to relevant parties. The purpose is supplemented by a justification for the research, and its scope.

2.2 Define objective(s)

The objective(s) of your research should illustrate what you wish to achieve through the research process. Objective(s) should be specific, measurable, achievable, realistic and useful.

2.3 Explain methodology

The methodology explains how you have planned to achieve the research objectives. The conceptual model will be presented together with the data needed, data sources, methods of data capture and analysis of data. In addition to these, the researcher should indicate the population, sample frame, sample size, sampling methods, and time plan for the study.

2.4 Develop your research proposal

This is your first major milestone. The supervisor will guide you in developing the proposal in the correct format. Often, a very strong proposal represents more than fifty percent of total work involved with the research project. What is a proposal? ... *A good proposal should consist of the first three chapters of the dissertation. It should begin with a statement of the research problem/background information, then move on to a review of the literature, and conclude with defining of the research methodology and should be written in future tense since it is a proposal.*

Before developing the proposal, question yourself assuming you have done a good job of "thinking about" your research project,

- whether you are familiar with other research carried out in related areas
- whether you have a clear understanding of the steps to use in conducting the research

- whether you are motivated, have confidence and the drive to go through each of the steps
- have read through someone else's research proposal
- have collected relevant articles and organized them methodically

The proposal should convince the examiners or the reader that you have come up with a unique problem. You should provide evidence to prove that there is a problem existing, and you have proposed a scientifically acceptable method of proposing a solution.

The format of the research proposal is given in Annex II.

2.5 Presenting the proposal and getting approval

A panel that includes the supervisor evaluates the research proposal. Your initial proposal is evaluated in terms of,

- significance and relevance of the issue and problem(s)
- conformity to format
- degree of reality, feasibility, importance of expected findings, and the possibility of using the findings to extend knowledge in the relevant area(s)

3. Carrying out the research

This involves conducting a literature review and/or exploratory study of the state of the art, modelling of a solution/concept/prototype, collecting primary and secondary data, analysing them, drawing conclusions, and reporting.

3.1 Relationship with the supervisor

- You should meet your supervisor *fortnightly* to discuss your research in detail.
- Seek out his/her input and assistance to guarantee that you are going along the correct path.
- Make sure your thought process and plan of action are cleared by the supervisor ahead of time before you go in front of the evaluation panel and discuss any specific strategy you should use at the defence.
- *Remember that in your final defence, your supervisor is your ally.* It is important that you detail out your presentation strategy with your supervisor in advance of your presentation, so that it is a team effort.

3.2 Literature survey

- Literature review can be used to *justify the need for your research, prove the originality and value of your contribution.* It presents readers a thorough review of the existing literature on the subject.
- Here you may also review the state of the art relevant to your research. The idea is to present the major ideas in the state of the art right up to but not including your own personal ideas.
- Literature review for the proposal may be narrow enough to prompt or to generate seed ideas, but a comprehensive literature review must be presented in your dissertation.

3.3 Methodology

You can either employ a quantitative approach or a qualitative approach. In many cases, a combined methodology makes the most sense. You may explain your overall research plan to express how you are going to execute the work to meet research objectives within the time available.

Acquiring *reliable* and *accurate* data that are relevant for your research project is an important determinant in achieving the research objectives. You may need specifying the method of analysing data. In addition to these, you should show sampling frame, sample size, sampling methods, and time plan of the study.

3.4 Sources and collection of data

Once you decide on the type of data - quantitative or qualitative - appropriate for the problem at hand, you will have to collect data through *observation, through a designed experiment, through a survey (by questioning) or from published sources*. You may need to employ one or more contact methods for collecting data.

The information you use should have integrity and support your research question. If you are using secondary sources as a source of data, then make sure to evaluate any secondary source for *point-of-view or bias, accuracy, date, authority and coverage* to ensure the interpretation is valid for your project.

3.5 Data analysis and data presentation

Data should be effectively presented, a simple rule if you are presenting information in the form of a table or graph make sure you introduce the table or graph in your text and then, following the insertion of the table/graph to discuss it. Label each table at the top, a picture or graph at the bottom. If there is nothing to discuss then you may want to question even inserting it. Where relevant, the accuracy of results should be presented quantitatively.

3.6 Making discussions, conclusions and recommendations

Conclusions are preceded necessarily by a discussion of your research findings. This is an objective and critical evaluation of solution alternatives that can be derived as your research outcomes. This should also include threats to validity of your work, comments on the reliability and accuracy of your results and limitations of your own work.

Conclusions are not a rambling summary of the dissertation but they should follow from the analysis and discussion: they are short, concise statements of the inferences that you have made from your work. Usually you can think of conclusions / implications as the "so what" statements. These may include recommendations for actions and further work.

3.7 Reporting research findings

Your research dissertation must essentially show the significance of the issue at hand, the research problems emerging out of the issue and the socio economic impact of the problem. The dissertation then shows already generated knowledge which can be disseminated in the form of research publication. The structure of final dissertation is given in Annex IV.

Annex I

Activity Schedule

Academic Year	Semester	Week	Activity
Current academic year	1	2	Submission of research interest / tentative topic
		4	Finalizing research topic and supervisor assignment
		10	Submission of research proposal
		11	Proposal presentation
	2	4	Submission of first three chapters of dissertation
		5	Progress review presentation
		Week before last week	Submission of preliminary soft-bound dissertation
		Last week	Final presentation
Next academic year	1	1	Submission of final hardbound dissertation
		2	Release of results

Annex II

Structure of Research Proposal

Use the following information for preparing your research proposal.

1. Title

Title of your research should reflect, imply, and point to the purpose and objective(s) of research. A sub-title may be added to further describe/qualify the title. You must avoid vague, generic, and ambiguous titles.

2. Background of research

This section shows the significance of the issue(s) addressed to in research, justification for the research problem and domain in the light of the current state of the art, the published work found in the literature and gaps.

3. Literature review

Literature review essentially builds up a “story” or a “case” for theoretical and application validation of your research. Quote the literature from which you found facts and figures that helped you to conceptualise and design the purpose and objective(s) of research.

4. Research Problem

In this section, you should clearly define the research problem that you are going to solve. It should also explain the purpose of research. The purpose is usually supplemented by a justification for the research, and its scope.

5. Objective(s) of research

This should show what you wish to achieve through the research process. Objective(s) of a research should be specific, realistic, achievable, and useful. This may also include research questions, sub-objectives and any hypotheses optionally.

6. Expected outcomes/deliverables

Expected outcomes/deliverables shows the benefits some segment could derive from your research.

7. Methodology

Explains how you are going to achieve the research objectives. You may also discuss types of data needed, data sources, and methods of gathering and analysing data to validate your work.

8. Time plan

This is a list of project activities showing time schedule. This time schedule should be realistic in terms your personal commitments.

9. References

It is very important to acknowledge others work that you have use to provide the essential background or context to your research proposal. All references should therefore be cited using the given format within the text of the proposal. Please see citations and referencing style given under the structure of dissertation in Annex IV.

Annex III

Evaluation Criteria

Milestone	Criterion	Allocated Marks %	Given Marks
Research Proposal & Proposal presentation (W1=20%)	Significance of research problem (purpose, objectives)	20	
	Literature review (justification of need, evidence of originality and value of contribution)	25	
	Methodology (overall plan, degree of feasibility)	20	
	Quality of research proposal (clarity, structure and formatting, proper use of citations & references)	25	
	Oral presentation of proposal (time management, clarity and defense)	10	
Total marks for Research proposal presentation (M1)		100	
First three chapters of dissertation & Progress review presentation (W2=25%)	Literature review (survey of the relevant literature, inclusion of key publications, understating of cited literature)	20	
	Methodology (novelty solution proposed / model formulated)	40	
	Quality of first three chapters (clarity, structure and formatting, proper use of citations & references)	20	
	Oral presentation of progress up to first three chapters (time management, clarity and defense)	10	
	Continuity of progress (keeping to deadlines, meetings with supervisor)	10	
Total marks for Progress review presentation (M2)		100	
Final dissertation & Final presentation (W3=40%)	Data collection and analysis (reliability of data and data gathering and validity of solution proposed / model formulated)	25	
	Findings, discussion, conclusions & future work (significance of findings, achievement of objectives, effectiveness of recommendations)	30	
	Quality of dissertation (clarity, structure, formatting, proper use of citations & references)	25	
	Oral presentation of dissertation (time management, clarity and defense)	10	
	Continuity of progress (keeping to deadlines, meetings with supervisor)	10	
Total marks for Final presentation (M3)		100	
Publication (W4=15%)	Conference full paper: 33 marks, indexed conference full paper: 66 marks, journal paper: 100 marks, more than one paper: marks for one paper + 33 bonus, indexed journal paper: 100 marks + 66 bonus	100	
Total marks for Publication (M4)		100	
Final Grade: (W1 × M1 + W2 × M2 + W3 × M3 + W4 × M4)		100	

Note 1: Students are required to obtain a minimum of 40% marks for each of the M1, M2 and M3 milestones in order to get through this course unit.

Note 2: To claim W4, students are required to get the full consent from his/her supervisor/s before submitting their papers to any conference/journal.

Annex IV

Structure of Dissertation

The structure and the general content requirements of your dissertation are described in this appendix.

1. Requirements

1.1 Margins, font and font sizes

Page margins top, right and bottom should be 25 mm. Left margin should be 38 mm to allow for binding. The text should use 1.5 line spacing and be fully justified. Recommended font is Times Roman size 12 point. Chapter headings should be 24 point bold. Level 1, 2 and 3 headings should be 18, 16 and 14 point bold, respectively. It is seldom necessary to have more than three levels of heading. However, if further subdivision is unavoidable, then do not number the level 4 heading, just highlight it in 12 point bold.

1.2 Page numbering and printing

All pages should be numbered with chapter 1 beginning on page 1. Use Roman letters for pages before that. Single-sided printing should be used for final report. Do not include title of research, author etc. in either the header or footer. All chapters should be begin on a new page.

1.3 Tables, figures and equations

All tables, figures and equations should be numbered with regard to chapter number and ordering within the chapter. Section numbers within the chapter do not influence this numbering. All tables and figures must include a caption which also appears in the list of figures and tables after the main table of contents. You should make sure to refer tables and figures by their number within the text.

1.4 Dissertation size

Your dissertation text (defined as everything except title page, table of contents, lists of tables and figures, references and appendices) should not normally exceed 120 pages except under special circumstances. On average it would be about 60-80 pages of A4 narrative text.

1.5 Writing style

Scientific writing style usually employs the passive form. It is considered not professional in scientific writing to make explicit references to what you yourself did, as in ‘I decided...’. Scientific papers rarely use the first person in this way. The passive form as in ‘it was decided...’ is usually preferred with ‘we’ used instead of

“I” in case of the use of active voice to indicate that a research work is never a sole effort.

1.6 Spelling and Grammar

You are required to get your dissertation proofread for any grammatical and/or spelling errors/mistakes and certified by a competent person to that effect. This should be done in the last page of soft bound version of your dissertation and should include the signature, name and designation of the certifier with date. It is your sole responsibility to ensure that there are no such errors in your dissertation.

However, note that this page should not be included in the final hardbound version of your dissertation.

1.7 Plagiarism

Plagiarism is the presentation of another person’s thoughts or words as though they were your own. You should explicitly avoid this when writing your dissertation. All sentences or passages quoted in your dissertation from other people’s work have to be specially acknowledged by clear cross-referencing. Direct quotations from published or unpublished work of others should always be clearly identified as such by being placed inside quotation marks or by being indented appropriately from both margins, and full reference to original source should be provided in the proper form. Equally, if another person’s ideas or judgements are summarised, you should refer to such work in the main text of the dissertation, and include the work referred to in the reference section. Failure to observe these rules may result in an allegation of cheating. Any illustrations, which are not your work, should also be acknowledged.

This research project is an important course unit of your degree and plagiarism in project work is taken very seriously will result in severe penalties. Therefore, it is important to give credit where it is due and emphasise your own contribution to the project.

2. Preface Materials

2.1 Title page

This comprises the title of your dissertation, your name, student number, the name(s) of the supervisor(s) and the following statement:

"This dissertation is submitted in partial fulfilment of the requirements for the degree of Bachelor of Science (Honours) in Management and Information Technology"

This should be followed by names of the department, faculty and university, country and the month and year of submission.

Note that the title should be chosen carefully so as to be not too long to be a sentence or paragraph and not too short to be unrepresentative of the contents of the dissertation.

2.2 Declaration

The second page should contain the following statement:

"I hereby state that this research report and findings presented in it are my own and it has not been submitted before nor is it currently being submitted for any other academic programme. Where material has been used from other sources due recognition has been given by mentioning the source."

You have to sign and date the above statement indicating the initial date of submission. Below your signature, your supervisor has a place to sign indicating the effective date of submission which will be the same as the initial date of submission if there have been only minor changes.

2.3 Abstract

The abstract should encourage the reader to find out more, and has to identify the main aspects and distinguishing characteristics of your research work. It should provide a clear statement of what your research is about, the scope of the project, a summary of the most important features/results set out in the dissertation, its relevance to the goals of the project and your achievements. The abstract should not be more than a single A4 page.

2.4 Acknowledgements

It is important to thank all who aided you in numerous ways during this serious piece of work. Acknowledgements should however be brief and to the point.

2.5 Contents

Provide page numbers of all sections, tables and figures. The overall structure should show a clear progression of logical thought. The structure of the dissertation should be seen to match the summary provided in the abstract.

2.6 Acronyms

You should provide all abbreviations used in the dissertation in alphabetical order with their expected versions.

3. Main Chapters

You are expected to introduce each chapter with its own introduction. This should summarize how the chapter fits into your dissertation, what the previous chapter asserts and indicate how this chapter follows it.

At the end of each chapter, you are also expected to summarize the contents of the chapter with its own summary. This should indicate how this chapter continues to the succeeding chapters.

3.1 Chapter 1: Introduction

This chapter should provide a general introduction to your research project with a clear description of its objectives. Initially, the case for the need of the research should be well argued. The subject area concerned is explained next, to provide the motivation for reading this dissertation. A detailed description of the problem context, history, impact, purpose and objectives of work should be provided. Your research findings should then be summarised in one or two paragraphs. Finally, introduce the structure of the dissertation (what will be covered in the remaining chapters).

3.2 Chapter 2: Literature Review

This chapter should include background material in the form of a literature review. What the reader should know in order to understand your work is provided here. A systematic and comprehensive review of relevant literature and other sources is documented here. A review of related work by other researchers and the relationship between your and their work should form the main thrust of this chapter. If several other researchers have done closely related work in a different way, the reasons for your approach should then be summarised here.

Your literature review should not simply an annotated list of papers you have read. It should cover a range of material relevant to your research work. Everything used should be cited by reference to the end of the dissertation.

3.3 Chapter 3 to n-1

You should provide your research methodology, how you collected data, analysed them and a critical self-evaluation of your own work in chapters from 3 to n-1. The type of research problem you are going to solve usually dictates the content and structure of these chapters. You should therefore discuss with your supervisor to determine a meaningful chapter wise classification of your research work. Note that here the research methodology should be written in past tense since, unlike in the proposal as it has now been completed.

3.4 Chapter n: Conclusions and future work

This chapter should present conclusions, further work and a summary of what have been achieved. A critical appraisal of your own work should also be included. Discuss the realisation of the original objectives/goals and how your work can be taken further, perhaps by another student in coming years. It should bring together many of the points mentioned in other chapters. Some of the earlier statements may be therefore repeated here.

3.5 References

It is very important to acknowledge others work that you have used or adapted in your research project, or that provides the essential background or context to dissertation.

All references should be cited using the given format within the text of your dissertation.

Citations should be described using author's name and year of publication in the following format:

- For one author: [Wijayanayake, 2010]
- For two authors [Wijayasiriwardhane and Lai, 2008]
- For more than two authors [Jayalal et. al., 2017]
- Lowercase letters, starting with 'a', are used to distinguish papers written by the same author during one calendar year, e.g. [Nanayakkara, 2016a]

In the references list, the references should be arranged in alphabetical order of author's name and within this in chronological order. The year follows the author's or authors' names. References should follow the format of the examples given below:

- Books
[Fonseka & Degamboda, 1992] Fonseka, M. and Degamboda, S., Introductory statistics for managers: a self-study manual, Postgraduate Institute of Management, 1992.
- Journal papers
[Peter et. al., 2010] Peter, P. L. S., de Bruijn, E. J. and Rwegasira, K., Performance of privatized enterprises and impact of partial privatization in a developing country: The case of Sri Lanka, *Transnational Management*, vol. 15, no. 1, pp. 46 – 68, 2010.
- Conference papers
[Higa & Wijayanayake, 1998] Higa, K. and Wijayanayake, J., Telework in Japan: perceptions and implementation, in *Proceedings of the Thirty-First Hawaii International Conference on System Sciences*, Kohala Coast, USA, pp. 169 – 177, January 1998.
- Theses/Dissertations
[Wijayasiriwardhane, 2010] Wijayasiriwardhane, H. K. T. K., Estimating system-level size and development effort of component-based systems using Component Point, vol. PhD. Melbourne, Australia: La Trobe University, 2010.

- Book chapters

[Jayawardena & Worboys, 1995] Jayawardena, D. P. W. and Worboys, M. F., The role of triangulation in spatial data handling, in *Innovations in Geographical Information Systems*, Fisher, P., Eds. Oxford: Taylor & Francis, pp. 7–18, 1995

- Web References

[Allen, 2002]

P. Allen, CBD survey: The state of the practice, *A white paper by the Cutter Consortium*,
<http://www.cutter.com/research/2002/edge020305.html>, 2002

Note that where an author name or date are not known, then the citation in the main text should take the form of a tag which clearly identifies the correct reference, e.g. for a World-Wide Web reference for which the author(s) is not known, the citation could take the following form (WWWn) where n are consecutive numbers.

- [WWW3]

<http://www.academiccolab.org/index.html>,
 accessed on 26/04/2017

3.6 Appendices

These may include further details of results, mathematical derivations, certain illustrative figures, user documentation and log of project milestones etc.

4. Submission of Dissertation

4.1 Preliminary version

This soft bound version of your dissertation will be used to examine your research work. Your supervisor will read this dissertation and may suggest improvements that you can incorporate in the final submission if he/she is satisfied with your work. Since this preliminary version is used to examine your project, it must be complete as possible.

4.2 Final version

You should incorporate suggestions given by your supervisor as much as possible. A separate document should be prepared describing how / why you incorporated / disregarded those suggestions and any other modifications done between the preliminary and final versions of your dissertations.

For final submission the dissertation must be hard bound in black Leather or Rexene in two copies with all text of the title page on outside cover in gold print. Spine must carry your name with initials, research title (abbreviated if necessary), degree and the year of submission in gold lettering of suitable size in the standard direction. One copy will be returned to you after certifying.

4.3 Sample title page

**A Method to Estimate the Functionality
Contributed by the Black-Box Type
Software Components in
Backfired Function Point**

Submitted by
Meera Devendran
IM/2007/025

Supervised by
Dr. Keerthi Wijayasiriwardhane

A dissertation submitted in partial fulfilment of the
requirements for the Degree of Bachelor of Science in
Management and Information Technology

**Department of Industrial Management
Faculty of Science
University of Kelaniya,
Sri Lanka
February 2013**